



TBScreen.AI

Setup & Run Guide

How to clone this project from GitHub and view the interface on your own device (browser or Android tablet).

AI-assisted TB screening · Flutter · Material 3 · UI-only prototype

1 What is this project?

TBScreen.AI is a **Flutter tablet application** for the **doctor role** of a tuberculosis (TB) screening workflow. A clinician can register a patient, capture or upload a chest X-ray, run an AI analysis, review and validate the result, manage training datasets, and sync the model — all from one tablet-optimised interface.

UI-only phase. There is no backend yet — every screen runs on built-in mock data and the AI / sync steps are simulated. You do **not** need any server, database, or API key to run it.

Target device

Final target is an **Android tablet in landscape** (10–12"). For a quick look you can also run it in a desktop **Chrome** browser — that is the fastest way to see the interface on any computer.

2 What you need first

Tool	Why	How to get it
Flutter SDK 3.x	Builds & runs the app (Dart 3.11+)	docs.flutter.dev/get-started/install
Git	Clones the project from GitHub	git-scm.com/downloads
Google Chrome	Fastest preview, no device needed	google.com/chrome
Android tablet / emulator	Optional — the real target device	Android Studio > Device Manager

Verify your setup after installing Flutter:

```
flutter --version
flutter doctor
```

3 Clone and run in 4 steps

Step 1 — Clone the repository from GitHub

```
# replace the URL with your repository address
git clone https://github.com/<your-username>/<your-repo>.git
cd <your-repo>
```

Step 2 — Download the dependencies

```
flutter pub get
```

Step 3a — Run in a browser (fastest, any computer)

```
flutter run -d chrome
```

Make the window wide. The tablet layout activates at **1024 px width or more**. Maximise the Chrome window. If a cramped layout appears, that is the responsive breakpoint — not a bug.

Step 3b — Run on an Android tablet or emulator (real target)

```
flutter devices # list connected devices, copy the id
flutter run -d <device> # e.g. flutter run -d emulator-5554
```

Step 4 — Explore the interface

The app opens on the **Login** screen. Enter **any valid-looking email** (e.g. doctor@tbscreen.app) and any password, then tap **Sign In**. You are taken to the Dashboard, and every screen is reachable from the navigation rail on the left. No real credentials are checked.

4 What you will see

Once signed in, these destinations are available from the left navigation rail:

Screen	What it shows
Dashboard	KPI cards, agreement bar, diagnosis-trend chart, case-distribution donuts.
Patients	Searchable patient list with profile, X-ray card and history timeline.
Diagnose	Patient intake form + X-ray upload/capture, then a simulated AI run.
Result	AI verdict, confidence, recommendations and summary (dark theme).
Validation	Review AI results case-by-case; agree / disagree with a note.
Dataset	Create, view, edit and manage training datasets and images.
Sync	Simulated model-update and consent-gated medical-data backup.
Account	Doctor profile, stats, security & notification settings, sign out.

Tip — primary flow. To experience the core journey: **Login** → **Diagnose** → **(upload X-ray)** → **Run AI Diagnosis** → **Result**. The Result page then shows the AI verdict in dark theme.

5 Troubleshooting

Symptom	Fix
'flutter' is not recognised	Add the Flutter bin folder to your PATH, then reopen the terminal.
The layout looks cramped / mobile	Widen the window to ≥ 1024 px. The design is tablet-landscape first.
Blank white/black page in browser	Wait for the first compile, then refresh once (F5).
'No devices found'	Run 'flutter doctor', start an emulator, or use '-d chrome'.
pub get fails	Check your internet connection and that the Flutter version is 3.x.

6 Good to know

- This is a **front-end prototype**: data is mock and AI/sync are simulated. No patient data leaves the device.
- A full **README.md** in the repository root documents the project structure and conventions for developers.
- Companion roles (**Admin RS**, **Super Admin**) are separate web repositories and are not part of this app.
- Build a release with `flutter build web` or `flutter build apk`.

Built with Flutter · Material 3 · UI-only prototype. Generated for the TBScreen.AI repository.